probability distribution. Let ϕ be a C^∞ function of the phase variables with compact support and define the time t average by

$$\bar{\phi}(t) = \frac{\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega}{\int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega}. \tag{5.6}$$

As the flow is Hamiltonian, $\int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega \equiv \int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},0)\mathrm{d}\omega$. Thus the condition for $\bar{\phi}$ to converge is just that

$$\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\frac{\partial\rho}{\partial t}(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega \to 0.$$

Since

$$\frac{\partial\rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

this suggests that the condition for the invariant probability density function ρ_∞ should be

$$\int_{\mathcal{D}} \phi \mathcal{L}_f^\dagger \rho_\infty \mathrm{d}\omega = 0,$$

for all suitable functions $\phi \in C_0^\infty(\mathbb{R}^m)$. We say that ρ_∞ satisfies

$$\mathcal{L}^\dagger \rho_\infty = 0,$$

in the *weak* or *variational* sense. When such a density exists, we refer to the associated probability distribution as an *invariant distribution*. Such invariant distributions are essential tools in molecular modelling. In the case of a Hamiltonian system, we know that $\mathcal{L}_f^\dagger = -\mathcal{L}_f$, and, moreover,

$$\mathcal{L}_f H = 0,$$

so that H is a first integral. Any probability distribution whose density is a smooth function of H will also be an invariant distribution.

In particular, consider the density

$$\rho_\varepsilon(\boldsymbol{q},\boldsymbol{p}) = Z_\varepsilon^{-1} e^{-(H(\boldsymbol{q},\boldsymbol{p})-\mathrm{E})^2/2\varepsilon},$$

where ε is an arbitrary small positive constant and E is an accessible energy level. We assume that H tends to infinity sufficiently rapidly as $\|\boldsymbol{z}\| \to \infty$ so that the exponential decays sufficiently rapidly (and thus the total integral exists) and take

Z_ε to be a normalization factor that insures that we have, for each ε, a well defined probability density function.

As we decrease ε, the density remains invariant under Liouville's equation. The prefactor means that we can think of a sequence of smooth densities taken with successively smaller values of ε as approaching the limiting distribution defined in terms of the delta function by

$$\rho_0(\boldsymbol{q},\boldsymbol{p}) = Z_{H=\mathrm{E}}^{-1}\delta[H(\boldsymbol{q},\boldsymbol{p}) - \mathrm{E}], \tag{5.7}$$

where $Z_{H=\mathrm{E}}$ is the appropriate limiting value of the normalization constant. To make this statement precise, the definition of the limit must be taken here in the weak sense, that is, we need to study the limit

$$\lim_{\varepsilon\to 0} Z_\varepsilon^{-1} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\varepsilon \mathrm{d}\omega, \tag{5.8}$$

taken in relation to a test function $\phi \in \mathscr{T}$. The proof that (5.8) can be taken and that

$$\lim_{\varepsilon\to 0} Z_\varepsilon^{-1} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\varepsilon \mathrm{d}\omega = \rho_0(\boldsymbol{q},\boldsymbol{p}) = Z_{H=\mathrm{E}}^{-1} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\delta[H(\boldsymbol{q},\boldsymbol{p}) - \mathrm{E}]\mathrm{d}\omega$$

may be found in the PhD thesis of Hartmann [171, Appendix D].

From a mathematical perspective, we may prefer to have a concept of the microcanonical probability measure that does not involve delta functions. It is possible to understand computations of averages with respect to a Dirac-type generalized function like ρ_0 as integrals on a set in a space of dimension one less than that of the ambient phase space (which is typically an even dimensional Euclidean space). We assume the level sets $\Sigma_\mathrm{E} = \{(\boldsymbol{q},\boldsymbol{p}) | H(\boldsymbol{q},\boldsymbol{p}) = \mathrm{E}\}$ to be bounded and define the invariant measure of a set $\mathcal{M} \subset \Sigma_\mathrm{E}$ by

$$\mu(\mathcal{M}) = \int_{\mathcal{M}} \frac{1}{\|\nabla H\|}\mathrm{d}\sigma,$$

where $\mathrm{d}\sigma$ is the "area element" of the energy surface. (The term "area" in this context might be replaced by "hyperarea" or "volume" since we are in many dimensions.) If we can represent the $m-1$-dimensional set $\mathcal{M}$ as the image of an open set $\mathcal{U} \in \mathbb{R}^{m-1}$ under a smooth mapping $\boldsymbol{g} : \mathbb{R}^{m-1} \to \Sigma_\mathrm{E}$, then we have

$$\mathrm{Vol}(\mathcal{M}) = \int_{\mathcal{U}} \sqrt{\det[(\boldsymbol{g}'(\boldsymbol{\zeta}))^T \boldsymbol{g}'(\boldsymbol{\zeta})]}\mathrm{d}\zeta_1\mathrm{d}\zeta_2\ldots\mathrm{d}\zeta_{m-1}.$$

Thus the invariant measure of interest to us is a rescaling of the indicated metric by the factor $1/\|\nabla H\|$.